Jawad Adit Chowdhury

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Professional Summary

Computer Science and Engineering graduate from BRAC University who builds full-stack web apps (React, Node.js, MongoDB) and applies deep learning to medical imaging (PyTorch, nnU-Net). Delivered three projects: a team-designed output head for nnU-Net v2 for pediatric brain tumor segmentation, evaluated with five-fold cross-validation on 257 MRI cases; a deployed AI subscription and finance platform built with a four-person team; and a sustainability tracker that generates personalised advice with the Gemini API. Experienced in designing REST APIs with authentication and ownership checks, building interactive dashboards with React and charting libraries, and running controlled ML experiments with cross-validation and statistical testing.

Projects

- **Automated 3D Segmentation of Pediatric Brain Tumors**
  PyTorch, nnU-Net v2, Python, BraTS-PEDs 2025
  - Worked with team members to design HSRD, a hybrid sigmoid-softmax output head for nnU-Net v2 that makes tumor sub-regions mutually exclusive via a shared softmax, ground-truth-masked loss, and an inference gate.
  - Evaluated against 3 baselines with five-fold cross-validation on 257 MRI cases using Dice, HD95, lesion-wise metrics, and paired Wilcoxon tests with Holm correction.
  - Removed sibling-label conflicts that appeared in 48 of 52 baseline validation cases and raised correctly empty ET-free predictions from 27 to 53 of 86 cases.
- **Stackwise: AI Subscription Optimizer & Personal Finance Tracker**
  React, Node.js, Express, MongoDB, Tailwind CSS, Gemini API
  - Built the subscription tracking module: a secured Express/MongoDB REST API (CRUD with per-user ownership checks) and a React drawer form with capability tags, trial alerts, and optional checkout.
  - Built a client/project profitability dashboard comparing revenue, costs, and margins per client, with Recharts category breakdowns.
  - Built a tax report generator with downloadable PDF export and a Gemini-powered scan that flags missed possible tax deductions.
  - Designed the subscription tracker screens in Figma. Deployed on Vercel (Lighthouse: 85 performance, 95 accessibility).
- **EcoTrack: Personal Sustainability Tracker**
  React, Node.js, Express, MongoDB, Chart.js, Gemini API
  - Built a MERN app (MVC) where users log eco-friendly and harmful activities to earn an Eco Score and a per-category carbon footprint estimate.
  - Implemented goals with progress tracking, badges and streaks, Chart.js trend and category charts, and daily reminders at a configurable time.
  - Integrated the Gemini API to generate personalised advice and suggested activities from each user's logged history.

Technical Skills

- **Programming Languages**: JavaScript, TypeScript, Python
- **Frontend**: React, Tailwind CSS, Bootstrap, Chart.js, Recharts
- **Backend & Data**: Node.js, Express.js, MongoDB, Mongoose, REST APIs
- **Machine Learning**: PyTorch, nnU-Net, medical image segmentation, statistical testing
- **Tools**: Git/GitHub, Postman, Figma, Vercel, pytest, draw.io
- **Spoken Languages**: English (Fluent), Bengali (Fluent), French (Basic)

Education

- **BRAC University** — Dhaka, Bangladesh
  Bachelor of Science in Computer Science and Engineering; CGPA: [3.18 / 4.00] — *2022 – 2026*